

Visible Cues: How Signifiers Shape Feature Discovery in a Smart-Thermostat App

A. Fixture, B. Synthetic. SYNTHETIC TEST PAPER written for the UXie test suite; the study, data and references are fictional.

Abstract

Many useful features of connected home devices are never found by the people who own them. We report a within-subjects lab study ($N = 24$) comparing three ways of presenting the same weekly-schedule feature in a smart-thermostat app: a hidden swipe gesture, an icon-only button, and a labelled button. Discovery within a five-minute exploration task rose from 21% (gesture) to 58% (icon) to 92% (label). Participants who found the feature but received no confirmation after saving a schedule often repeated the action. We argue that a feature can be possible without being perceivable, and that designers must separately plan the cue that signals an action and the feedback that confirms it.

1 Introduction

Owners of smart thermostats routinely say that their device 'cannot' do things it can in fact do. Support forums are full of requests for scheduling, a feature that most of these apps already offer. This paper asks why such features stay invisible and what interface cues make them discoverable. Our contribution is a controlled comparison of three cue designs for one feature, plus qualitative observations of how people search an unfamiliar app.

2 Background

2.1 Affordances and signifiers

In interaction design, an affordance describes a relationship between an object and an agent: what actions are possible for this agent with this object. A touchscreen region that responds to a swipe affords swiping, whether or not anyone can tell. A signifier is different. It is a perceivable cue, such as a label, an icon, a shadow or an arrow, that communicates where an action is possible and how to perform it. Designers often blur the two terms. We keep them apart because our three conditions hold the affordance constant (the schedule feature is equally possible in all of them) while varying only the signifier.

2.2 Feedback and the gulf of execution

The gulf of execution is the gap between what a person wants to do and what the system lets them do in a way they can see. Signifiers help bridge it before an action. Feedback works after the action: it tells the person what happened, so they can judge whether the goal was reached (bridging the gulf of evaluation). Prior lab work on mobile settings screens suggests that missing feedback leads to repeated actions and lower confidence, but those studies did not vary the signifier at the same time.

2.3 Hidden gestures

Gesture-only controls save screen space and look clean, which explains their popularity in minimalist home-automation apps. Their cost is that nothing on screen signals them.

3 Method

3.1 Participants

We recruited 24 participants (13 women, 11 men; aged 19 to 41, median 24) from a university mailing list. Eighteen were students. None had used the test app before; nine owned some smart-home device.

3.2 Design and materials

We built three versions of a thermostat app prototype that differed only in how the weekly-schedule feature was presented on the home screen: (G) Gesture: swiping left on the temperature dial opened the schedule; nothing on screen indicated this. (I) Icon: a calendar icon without text in the top-right corner. (L) Label: a button reading 'Weekly schedule' with the same icon. Each participant used all three versions in counterbalanced order, each with a different visual theme so that learning from one version did not carry over directly. After saving a schedule, half of the sessions in each condition showed a confirmation toast ('Schedule saved') and half showed nothing.

3.3 Tasks and measures

Participants first explored each version freely for five minutes while thinking aloud, then received the task 'Make the heating come on at 6:30 on weekdays'. We measured (1) discovery: whether the schedule screen was opened during free exploration; (2) time to first use once the task was given; (3) repeated saves, as a sign of uncertainty; and (4) a 7-point confidence rating after the task. Sessions were video recorded and two researchers coded the think-aloud transcripts.

4 Results

4.1 Discovery

During free exploration, 5 of 24 participants (21%) discovered the schedule in the Gesture version, 14 of 24 (58%) in the Icon version and 22 of 24 (92%) in the Label version. The difference between Gesture and Label was significant (McNemar test, $p < .001$); Icon versus Label was also significant ($p = .02$).

4.2 Time to first use

Once given the task, median time to open the schedule was 74 s for Gesture, 19 s for Icon and 6 s for Label. Seven participants never found the gesture and were shown it after three minutes.

4.3 Feedback

Without a confirmation toast, 11 of 36 schedule saves (31%) were repeated at least once; with the toast, 2 of 36 (6%) were. Mean confidence was 4.1 without and 6.0 with the toast. The toast had a similar effect in all three cue conditions.

4.4 Think-aloud themes

Participants searched the screen for 'something that looks tappable' and rarely tried gestures unless they had a reason to. Several said the icon could mean 'calendar', 'history' or 'holidays'. Two participants who found the gesture by accident could not repeat it later. Participants with smart-home experience were not faster at finding the gesture.

5 Discussion

Our results separate two questions that designers often merge. Whether an action is possible (the affordance) was identical across conditions, yet discovery ranged from one in five to nearly everyone. What changed was the signifier. A hidden gesture offers no cue at all; an icon offers a cue whose meaning is ambiguous; a label offers a cue that names the action.

Feedback addressed a different problem. Even participants who found the feature were unsure whether saving had worked when nothing confirmed it, and they repeated the action. Signifiers and feedback therefore play complementary roles: one helps people find and start an action, the other helps them judge its result.

5.1 Implications for design

(a) When a feature matters, give it a perceivable signifier rather than relying on gestures alone; a gesture can remain as a shortcut. (b) Icons should be paired with text when their meaning is not conventional. (c) Every state-changing action needs feedback that names the new state. (d) Labels compete for space: adding labels to every control can crowd a screen, so prioritise the features users most often fail to find.

6 Limitations

This was a short lab study with 24 mostly young, university-educated participants and one app. Free exploration for five minutes is not the same as months of everyday use, where people might eventually discover gestures through tips or friends. We tested one feature and one gesture; other gestures may be more guessable. The confirmation toast was the only form of feedback we varied. Our discovery measure counts any opening of the schedule screen, including accidental ones. We did not measure long-term retention, and we did not study older adults or people using assistive technology, for whom hidden gestures may be even harder.

7 Conclusion

A feature that people cannot see is, for practical purposes, a feature they do not have. In our study, changing only the visible cue for a weekly-schedule feature raised discovery from 21% to 92%, and adding a one-line confirmation reduced repeated actions fivefold. Designers should plan the signifier that invites an action and the feedback that confirms it as two separate decisions.

References

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